

SHIVANSH TIWARI

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SUMMARY

Python GenAI Engineer with 1 year of production experience building RAG pipelines, multi-agent LLM systems, and LLM-integrated backends. Skilled in LangGraph, LangChain, vector search (ChromaDB, FAISS), and prompt engineering with direct OpenAI / Anthropic / Gemini API integration. Delivered 6+ quantified outcomes including 25% review-time reduction and 7s query-latency drop, Strong foundation in Python, FastAPI, PySpark, and cloud deployment on Azure and AWS.

SKILLS

AI & GenAI: LangGraph, LangChain, RAG, OpenAI API, Anthropic Claude, Prompt Engineering, Vector Search, ragas, LangSmith, DeepEval

Languages & Databases: Python, C++, SQL, MongoDB, ChromaDB, FAISS

Backend & Web: FastAPI, REST APIs, Node.js, React.js, MCP (Model Context Protocol)

Data & Analytics: PySpark, Pandas, NumPy

Cloud & Tools: Azure, AWS (EC2, S3), Docker, Git/GitHub, Hugging Face

EXPERIENCE

Programmer Analyst Trainee

Oct 2025 – Present

Cognizant Technology Solutions

Noida · Hybrid

- Built a Python & PySpark scanning tool that sampled and processed 100M+ MS SQL records, enabling enterprise-scale Master Data Management (MDM) cataloging.
- Integrated GPT-5 mini and text-embedding-3-large via direct OpenAI API (bypassing LangChain for token-budget control) to auto-generate data dictionaries, categories, and domains with structured JSON output schemas.
- Developed a scalable FastAPI backend on Azure Database for PostgreSQL to store and manage catalog metadata, serving 500+ concurrent catalog read requests with <200ms p95 latency.
- Automated database schema reading and contextual tagging with semantic chunking (512-token overlap) and evaluated retrieval quality using ragas; improved answer relevance from 0.62 to 0.84.
- Engineered zero-shot and few-shot prompts with strict context boundaries and OWASP-aligned prompt injection defenses, constraining LLM outputs to internal documents only.
- Implemented LangSmith tracing for the cataloging pipeline, enabling real-time hallucination detection and reducing false-positive tagging by 18%.

Software Engineering Intern

May 2025 – Sep 2025

Cognizant Technology Solutions

Bangalore · On-site

- Prototyped a document summarization tool using direct OpenAI API with custom retry logic and token-budget enforcement, reducing internal report review time by 25%.
- Supported the core GenAI engineering team with prompt iteration, output quality assurance via DeepEval (faithfulness, answer relevancy), and feature documentation.

PROJECTS

DataOps Agent — Multi-Agent Data Analyst

[Link](#)

LangGraph, Gemini 2.5 Flash, DuckDB, Python, FastAPI

- Built a 6-agent system using LangGraph & Gemini 2.5 Flash for automated Text-to-SQL generation and dynamic charting with human-in-the-loop approval gates for low-confidence queries.
- Designed OWASP-aligned prompt injection defenses and input sanitization layers, blocking 94% of adversarial test prompts in red-team evaluation.
- Integrated in-process DuckDB and Anthropic MCP (Model Context Protocol), reducing query execution latency by ~7 seconds versus traditional SQL connectors
- Deployed on Render with Docker containerization also added LangSmith tracing for agent-step observability and debugging.

PyMind — Enterprise Python RAG Assistant

[Link](#)

LangGraph, FastAPI, Gemini LLM, ChromaDB, Turso Cloud, Hugging Face

- Architected a LangGraph & FastAPI state machine with early-exit routing that reduces out-of-topic query latency to ~1ms.
- Combined ChromaDB vector search (all-MiniLM-L6-v2 embeddings, 384-dim) and BM25 keyword search across 600K+ Stack Overflow chunks with an on-the-fly Tavily web fallback when local context is missing.
- Cut cloud database transaction latency from 5s to <50ms on Turso Cloud using a singleton connection cache and connection pooling.

EDUCATION

B.Tech, Computer Science & Engineering

2021 – 2025

SRM Institute of Science and Technology, Ghaziabad, UP · CGPA: 9.07 / 10

CERTIFICATIONS & AWARDS

- GenAI & Data Engineering Achievement Award — Cognizant
- Generative AI Fundamentals — Databricks
- AWS Cloud Foundations — AWS Academy